

WILKERSON PASS, CO, USA

WMO: 726396

Lat: 39.050N

Lon: 105.516W

Elev: 3438

StdP: 66.29

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 00422

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.9	(c) -17.0	(d) -35.9	(e) 0.2	(f) -1.8	(g) -31.0	(h) 0.3	(i) -0.9	(j) 11.0	(k) -4.3	(l) 10.1	(m) -5.3	(n) 4.5	(o) 320	(p) 0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 22.6	(d) 7.4	(e) 21.2	(f) 7.3	(g) 19.7	(h) 6.9	(i) 10.8	(j) 16.3	(k) 10.0	(l) 15.9	(m) 9.2	(n) 15.1	(o) 2.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	10.8	12.6	7.7	10.0	11.2	7.2	9.7	10.8	41.8	16.6	39.4	15.7	37.0	14.9	14.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.4	7.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-22.3	12.3	1.9	0.9	-23.7	12.9	-24.8	13.5	-25.9	14.0	-27.3	14.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-6.0	-5.3	-1.8	1.1	5.2	12.0	14.5	12.9	10.0	4.7	-0.6	-5.3		
	DBStd	8.34	4.54	5.04	4.55	4.17	4.29	3.61	2.72	2.68	3.45	4.59	5.04	4.87		
	HDD10.0	2743	496	429	364	267	158	21	3	7	38	168	317	474		
	HDD18.3	5418	754	663	623	517	408	189	122	169	249	423	567	733		
	CDD10.0	372	0	0	0	0	8	83	142	97	39	4	0	0		
	CDD18.3	4	0	0	0	0	0	1	2	1	0	0	0	0		
	CDH23.3	24	0	0	0	0	0	6	13	6	0	0	0	0		
	CDH26.7	1	0	0	0	0	0	0	1	1	0	0	0	0		
(14) Wind	WSAvg	3.7	4.3	4.6	4.0	3.8	3.2	3.1	2.6	2.6	3.2	3.7	4.4	4.5		
(15) Precipitation	PrecAvg	466	20	23	40	56	53	40	60	61	41	29	24	20		
	PrecMax	625	48	56	119	126	153	78	114	111	195	65	61	61		
	PrecMin	304	8	3	0	8	21	6	13	25	1	4	8	4		
	PrecStd	76	9	12	25	30	29	22	29	20	34	16	12	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.4	11.9	14.1	18.8	24.0	25.0	23.8	20.9	17.3	12.3	6.0		
		MCWB	-2.4	-1.5	0.5	0.1	4.1	6.0	7.7	8.5	6.0	3.9	0.0	-2.6		
	2%	DB	2.6	4.1	8.8	12.1	16.2	22.2	22.8	21.3	18.6	14.1	9.1	2.8		
		MCWB	-3.7	-2.9	-0.5	0.4	3.7	6.1	7.6	8.4	5.9	2.7	-0.3	-3.6		
	5%	DB	1.2	2.4	7.0	9.1	13.7	20.2	21.4	19.1	17.4	12.5	7.4	1.3		
		MCWB	-4.2	-3.6	-1.4	0.2	3.0	5.8	7.9	7.8	5.5	2.5	-1.1	-4.2		
	10%	DB	-0.1	0.9	4.8	7.3	12.0	18.0	19.8	17.6	16.0	11.0	5.2	0.1		
		MCWB	-4.7	-4.2	-2.4	-0.7	2.9	5.6	7.6	7.5	5.3	2.0	-2.2	-4.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.4	-0.1	1.3	3.0	6.4	10.0	12.2	12.4	9.7	6.7	1.3	-1.1		
		MCDB	2.7	5.5	8.2	10.0	13.6	16.5	17.3	18.2	15.8	12.7	8.0	3.2		
	2%	WB	-2.7	-2.0	0.1	1.6	5.2	8.7	10.9	10.8	8.4	5.4	0.2	-2.3		
		MCDB	1.3	2.7	7.2	7.5	12.2	15.4	16.6	15.9	14.0	11.1	6.9	1.2		
	5%	WB	-3.7	-2.9	-0.9	0.6	4.1	7.9	10.1	10.1	7.4	4.0	-0.6	-3.3		
		MCDB	0.3	1.4	5.3	6.5	10.9	14.6	16.4	15.3	12.9	9.1	5.8	0.1		
	10%	WB	-4.6	-3.9	-1.9	-0.1	3.5	7.0	9.4	9.2	6.5	2.9	-1.5	-4.2		
		MCDB	-0.8	0.0	3.4	5.7	10.0	14.1	15.8	14.5	12.7	8.1	4.5	-0.7		
(35) Mean Daily Temperature Range	MDBR		8.0	8.9	9.8	10.3	10.2	11.8	10.7	10.5	10.9	9.7	8.6	7.4		
	5% DB	MCDBR	7.7	9.3	10.7	12.2	12.5	13.5	12.1	12.1	11.9	11.2	9.4	7.4		
		MCWBR	4.9	5.6	5.5	5.6	5.0	4.8	4.0	4.3	4.7	5.1	4.8	4.5		
	5% WB	MCDBR	7.7	8.4	10.0	10.3	11.4	12.0	10.6	10.4	10.4	10.2	8.9	7.3		
		MCWBR	5.3	5.6	5.5	5.2	5.1	5.0	4.6	4.6	4.7	4.9	4.9	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.193	0.197	0.216	0.228	0.243	0.254	0.275	0.269	0.242	0.221	0.202	0.194		
	taud		2.697	2.703	2.662	2.671	2.693	2.666	2.641	2.638	2.732	2.750	2.732	2.691		
	Ebn at Noon		1043	1072	1066	1058	1037	1020	996	1000	1021	1026	1024	1020		
	Edh at Noon		61	70	82	87	87	90	92	89	76	66	58	57		
(44) All-Sky Solar Radiation	RadAvg		2.83	3.91	5.05	5.80	6.27	7.03	6.29	5.74	5.22	4.01	2.93	2.42		
	RadStd		0.11	0.20	0.40	0.39	0.49	0.39	0.32	0.28	0.31	0.24	0.16	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (31 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	+19

Nomenclature: See separate page